Problem 5.15: Meson Color Singlet Wavefunction

Griffiths, *Introduction to Elementary Particles*

A Pedagogical Solution

Problem Statement

5.15: Construct the (singlet) color wave function for mesons, analogous to Equation 5.60.

Equation 5.60 (the baryon color singlet):

$$\psi(\text{color}) = \frac{1}{\sqrt{6}}(rgb - rbg + gbr - grb + brg - bgr) \tag{5.60}$$

1 Setting the Scene: What's Different About Mesons?

You've already seen the baryon color singlet (Eq. 5.60). That involved three quarks, each carrying one of the three colors, combined in a fully antisymmetric way. Now we need to do the analogous thing for mesons — but the structure is fundamentally different.

Baryons are made of three quarks: qqq. Each quark carries a **color**: r, g, or b. The color singlet requires a fully antisymmetric combination of all three colors (the Levi-Civita symbol ϵ_{ijk}). This is Eq. 5.60.

Mesons are made of a quark and an antiquark: $q\bar{q}$. The quark carries a **color** (r, g, or b) and the antiquark carries an **anticolor** ($\bar{r}$, $\bar{g}$, or $\bar{b}$).

So the question is: how do we combine color + anticolor to get a **color singlet** (i.e., a state with no net color)?

A color singlet is a state that is **invariant under SU(3) color transformations**. Physically, this means it has *no net color charge* — it's "white." This is the requirement of **confinement**: only color singlets can exist as free particles.
For baryons, you get "white" by mixing all three primary colors ($r+g+b$) antisymmetrically. For mesons, you get "white" by pairing each color with its anticolor — like mixing a color with its complement.

2 Building the Meson Color Singlet

2.1 Step 1: List All Possible Color–Anticolor Combinations

A quark can be r, g, or b. An antiquark can be $\bar{r}$, $\bar{g}$, or $\bar{b}$. So there are $3 \times 3 = 9$ possible color–anticolor combinations:

$$\begin{array}{ccc} r\bar{r} & r\bar{g} & r\bar{b} \\ g\bar{r} & g\bar{g} & g\bar{b} \\ b\bar{r} & b\bar{g} & b\bar{b} \end{array}$$

In the language of group theory, this is the tensor product of the fundamental representation **3** (quark) and the conjugate representation $\bar{\mathbf{3}}$ (antiquark):

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}$$

The 9 combinations decompose into an **octet** (8 states) and a **singlet** (1 state). The singlet is what we want.

2.2 Step 2: Use the Analogy with Spin

This is where your intuition from spin addition can help enormously. The math of combining a color **3** with an anticolor $\bar{\mathbf{3}}$ is closely analogous to combining two angular momenta.

Recall from spin-$\frac{1}{2}$ addition: two spin-$\frac{1}{2}$ particles combine as:

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{3} \oplus \mathbf{1}$$

The singlet (total spin $S = 0$) is the *symmetric equal-weight superposition with alternating signs*:

$$|0,0\rangle = \frac{1}{\sqrt{2}}\left(\uparrow\downarrow \; - \; \downarrow\uparrow\right)$$

For color SU(3), the singlet in $\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}$ follows the same logic, but now with three basis states instead of two: the singlet is the *equal-weight superposition over all color–anticolor diagonal pairs.*

2.3 Step 3: Identify the Singlet

The singlet must be invariant under *any* SU(3) color rotation. The only combination of the 9 states that has this property is the one that treats all three colors **democratically** — an equal superposition of each color paired with its own anticolor:

$$\boxed{\psi(\text{color}) = \frac{1}{\sqrt{3}}\left(r\bar{r} + g\bar{g} + b\bar{b}\right)}$$

This is the answer to Problem 5.15.

3 Why This is the Answer: A Careful Explanation

3.1 Why Equal Coefficients?

The singlet must look the same no matter how you "rotate" in color space (i.e., relabel the colors). If we swapped the names $r \leftrightarrow g$, the state must not change. The only way to achieve this is if all three terms $r\bar{r}$, $g\bar{g}$, $b\bar{b}$ enter with **equal weight**.

3.2 Why All Plus Signs (Unlike the Baryon)?

This is a common point of confusion. The baryon singlet (Eq. 5.60) has alternating signs because it involves the **Levi-Civita symbol** ϵ_{ijk} — the fully *antisymmetric* combination of three objects. That's because baryons have three quarks (all in the fundamental **3** representation), and the singlet in $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$ is antisymmetric.

For mesons, we have a **3** and a $\bar{\mathbf{3}}$. The singlet in $\mathbf{3} \otimes \bar{\mathbf{3}}$ is built with the **Kronecker delta** $\delta_{i\bar{j}}$, which is *symmetric*. Explicitly:

$$\psi = \sum_{i=1}^{3} \delta_{i\bar{i}} \ |c_i\bar{c}_i\rangle = |r\bar{r}\rangle + |g\bar{g}\rangle + |b\bar{b}\rangle$$

No minus signs. The Kronecker delta has all $+1$'s on the diagonal.

	Baryon (qqq)	**Meson** $(q\bar{q})$
Representation	$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$	$\mathbf{3} \otimes \bar{\mathbf{3}}$
Singlet built from	ϵ_{ijk} (Levi-Civita)	$\delta_{i\bar{j}}$ (Kronecker delta)
Number of terms	6	3
Signs	Alternating $(+/-)$	All positive
Normalization	$1/\sqrt{6}$	$1/\sqrt{3}$

3.3 Why $1/\sqrt{3}$?

The normalization is straightforward. We need $\langle\psi|\psi|\psi|\psi\rangle = 1$. Since the three basis states are orthonormal:

$$\langle\psi|\psi|\psi|\psi\rangle = \frac{1}{3}\left(|\langle r\bar{r}|r\bar{r}|r\bar{r}\rangle|^2 + |\langle g\bar{g}|g\bar{g}|g\bar{g}\rangle|^2 + |\langle b\bar{b}|b\bar{b}|b\bar{b}\rangle|^2\right) = \frac{1}{3}(1{+}1{+}1) = 1 \quad \checkmark$$

The cross terms vanish because $\langle r\bar{r}|g\bar{g}|r\bar{r}|g\bar{g}\rangle = 0$, etc.

4 What About the Other 8 States?

You might wonder: we started with 9 combinations but the singlet uses only the 3 diagonal ones. What happened to the other 6?

The remaining 8 states form the **color octet** (**8**). These are:

The 6 "off-diagonal" states (these carry net color, like a gluon):

$$r\bar{g}, \quad r\bar{b}, \quad g\bar{r}, \quad g\bar{b}, \quad b\bar{r}, \quad b\bar{g}$$

Plus 2 "diagonal" combinations orthogonal to the singlet:

$$\frac{1}{\sqrt{2}}(r\bar{r} - g\bar{g}), \qquad \frac{1}{\sqrt{6}}(r\bar{r} + g\bar{g} - 2b\bar{b})$$

These 8 states are exactly the color states of the **8 gluons**. Gluons carry color–anticolor combinations, but specifically the octet combinations, not the singlet. This is why there are 8 gluons, not 9.

If a singlet gluon $(r\bar{r}+g\bar{g}+b\bar{b})/\sqrt{3}$ existed, it would be colorless and could propagate freely between hadrons like a photon. This would create a long-range strong force, which is not observed. The absence of the 9th gluon (the color singlet gluon) is related to the fact that the strong force is **confining** at long distances.

5 Verifying SU(3) Invariance

For the mathematically inclined, here's a sketch of why $(r\bar{r} + g\bar{g} + b\bar{b})/\sqrt{3}$ is invariant under SU(3).

Under an SU(3) color transformation U, a quark color state transforms as:

$$|c_i\rangle \to U_{ij}\,|c_j\rangle$$

An antiquark transforms with the conjugate:

$$|\bar{c}_i\rangle \to U^*_{ij}\,|\bar{c}_j\rangle$$

The singlet state $\psi = \sum_i |c_i\bar{c}_i\rangle$ transforms as:

$$\psi \to \sum_i U_{ij}\,U^*_{ik}\;|c_j\bar{c}_k\rangle = \sum_{jk}\left(\sum_i U_{ij}\,U^*_{ik}\right)|c_j\bar{c}_k\rangle$$

Since U is unitary, $\sum_i U_{ij}U^*_{ik} = (U^\dagger U)_{kj} = \delta_{kj}$. So:

$$\psi \to \sum_j |c_j\bar{c}_j\rangle = \psi$$

The state is invariant under *any* SU(3) transformation. It is a true singlet. □

6 Physical Meaning

The meson color wavefunction

$$\psi(\text{color}) = \frac{1}{\sqrt{3}}\left(r\bar{r} + g\bar{g} + b\bar{b}\right)$$

tells us that inside a meson like the π^+ $(u\bar{d})$, the quark and antiquark are in a **superposition** of all three color–anticolor pairings simultaneously:

- With probability 1/3: the u is red and the $\bar{d}$ is anti-red
- With probability 1/3: the u is green and the $\bar{d}$ is anti-green
- With probability 1/3: the u is blue and the $\bar{d}$ is anti-blue

At any instant, the quark and antiquark always have matching color–anticolor, so the total color charge is zero — the meson is "white." The specific color fluctuates quantum mechanically, but the net color is always zero.

This is analogous to how a spin singlet $(\uparrow\downarrow - \downarrow\uparrow)/\sqrt{2}$ has zero total spin even though the individual spins are fluctuating.

7 Complete Summary

The color singlet wavefunction for mesons is:

$$\boxed{\psi(\text{color}) = \frac{1}{\sqrt{3}}\left(r\bar{r} + g\bar{g} + b\bar{b}\right)}$$

	Baryon singlet (Eq. 5.60)	**Meson singlet (this problem)**
Structure	qqq	$q\bar{q}$
Invariant tensor	ϵ_{ijk}	$\delta_{i\bar{j}}$
Wavefunction	$\frac{rgb - rbg + \cdots}{\sqrt{6}}$	$\frac{r\bar{r} + g\bar{g} + b\bar{b}}{\sqrt{3}}$
# of terms	6	3
Signs	Alternating	All positive